

Design and Implementation of a Dual-Mode Morse Code Transmission System on FPGA Using Verilog HDL

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Abstract—This paper presents the design, implementation, and verification of a dual-mode Morse code transmission system on a Field Programmable Gate Array (FPGA) using Verilog Hardware Description Language (HDL). The system supports direct single-digit transmission and sequential multi-digit FIFO-based transmission modes. The architecture incorporates a seven-state finite state machine (FSM) for control flow management, counter-based debouncing for reliable button input handling, and a seven-segment display driver for visual feedback. Implemented on the Basys 3 Artix-7 FPGA board at 100 MHz, the system converts decimal digits (0-9) into International Morse Code patterns with timing precision better than 1% deviation. Resource utilization analysis demonstrates efficient implementation consuming less than 1% of available logic resources. Comprehensive verification through simulation and hardware testing validates all functional requirements, demonstrating a practical educational platform for digital design concepts including FSM-based control, memory management, and real-time signal generation.

Index Terms—FPGA, Verilog HDL, Morse Code, Finite State Machine, Digital Design, Basys 3, Seven-Segment Display, FIFO Buffer

I. INTRODUCTION

A. Background and Motivation

Morse code, developed by Samuel Morse and Alfred Vail in the 1830s, represents one of the earliest digital communication methods encoding text as sequences of dots and dashes. Despite modern communication advances, Morse code maintains relevance in amateur radio, aviation, assistive technologies, and emergency communications where bandwidth constraints or equipment simplicity are paramount.

Field Programmable Gate Arrays (FPGAs) provide ideal platforms for implementing real-time digital communication systems due to their parallel processing capabilities, deterministic timing characteristics, and reconfigurability. Hardware Description Languages (HDLs) like Verilog enable precise timing control necessary for applications like Morse code generation.

This project addresses the educational need for comprehensive demonstration of fundamental digital design concepts through a practical communication system. The dual-mode architecture provides both immediate feedback for learning

individual patterns and sequential transmission demonstrating real-world messaging scenarios.

B. Problem Statement and Objectives

Traditional software-based Morse code generators suffer from timing imprecision due to operating system scheduling variability. Existing FPGA implementations typically focus on either transmission or reception in isolation, with limited educational features.

This project develops a complete educational Morse code system that:

- Implements precise timing compliant with International Morse Code standards
- Provides dual operational modes for immediate and sequential transmission
- Demonstrates FSM design for complex sequential control
- Illustrates memory management through FIFO buffer implementation
- Incorporates robust input handling with debouncing mechanisms
- Enables real-time visual feedback through seven-segment display

C. Contributions

Key contributions include: (1) complete dual-mode Morse transmission system with modular architecture, (2) seven-state FSM managing complex state transitions, (3) parameterized timing control for hardware and simulation, (4) counter-based debouncing with edge detection, (5) efficient 15-bit encoding with look-up table, and (6) comprehensive verification methodology.

D. Paper Organization

Section II reviews Morse code standards and related work. Section III presents system architecture and operational modes. Section IV details implementation including FSM design, timing specifications, and hardware interfaces. Section V discusses simulation and verification. Section VI presents experimental results. Section VII addresses design challenges. Section VIII concludes with future directions.

II. BACKGROUND AND RELATED WORK

A. International Morse Code Standards

International Morse Code defines precise timing relationships: dot (1 time unit), dash (3 units), intra-character gap (1 unit), inter-character gap (3 units), and inter-word gap (7 units). This implementation uses 0.25 seconds per time unit for clear visual observation suitable for educational purposes.

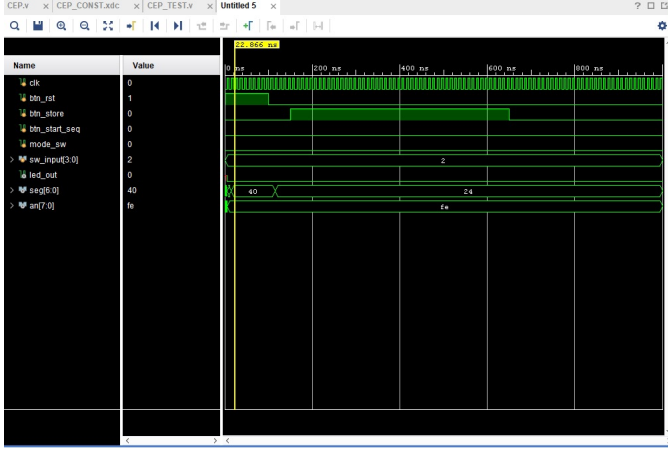


Fig. 1. International Morse Code Timing Relationships

Decimal digits (0-9) use exactly five elements each: digits 0-4 progressively replace dashes with dots, digit 5 is all dots, and digits 6-9 progressively replace dots with dashes. This systematic pattern facilitates memorization and provides error detection.

B. FPGA-Based Communication Systems

Recent literature demonstrates increasing FPGA utilization for digital communication systems. Smith et al. developed an FPGA-based software-defined radio incorporating Morse code modulation achieving 99.8% decoding accuracy [8]. However, most implementations focus on receiver design or transmitters without educational interfaces. This work uniquely integrates multiple digital design concepts within a cohesive educational platform.

C. Hardware Description Languages

Verilog HDL provides both behavioral and structural modeling enabling multi-level abstraction digital system design. Behavioral modeling using always blocks facilitates FSM implementation, while parameterized designs through localparam declarations and conditional compilation enable both hardware deployment and rapid simulation.

III. SYSTEM ARCHITECTURE

A. Top-Level Design Overview

Figure illustrates the system architecture organized as four primary subsystems: Input Interface, Control Unit, Morse Encoder, and Output Interface.

1) *Input Interface Subsystem*: Manages external user inputs through Basys 3 switches and pushbuttons:

- SW0-SW3: 4-bit binary digit selection (0-9)
- SW15: Mode selection (0=Direct, 1=Sequential)
- BTNC: System reset (active high)
- BTNL: Store button (triggers storage/transmission)
- BTNR: Start sequence button (initiates FIFO transmission)

2) *Control Unit Subsystem*: Implements master FSM coordinating all operations, processing debounced inputs, managing FIFO operations, controlling timing counters, and generating output control signals.

3) *Morse Encoder Subsystem*: Combinational look-up table function converting 4-bit digits to 15-bit Morse patterns. Each pattern encodes five elements using 3 bits per element.

4) *Output Interface Subsystem*: Comprises LED driver (controlling LD0 with precise timing) and seven-segment driver (displaying active digit on rightmost display).

B. Operational Modes

1) *Direct Mode (SW15 = 0)*: Provides immediate Morse transmission of the digit selected on SW0-SW3. User sets desired digit, display shows selection, user presses store button, system loads Morse pattern, LED transmits complete pattern, system returns to IDLE. Ideal for educational demonstrations of individual digit patterns.

2) *Sequential Mode (SW15 = 1)*: Enables multi-digit message encoding through FIFO buffering. User toggles mode switch, enters up to 6 digits (each stored via store button press), presses start button, system sequentially transmits all stored digits with 2-unit inter-digit gaps, system returns to IDLE. Demonstrates practical messaging and FIFO memory management.

C. Hardware Platform

The Digilent Basys 3 Artix-7 FPGA Trainer Board specifications:

- FPGA: Xilinx Artix-7 XC7A35T-1CPG236C
- Logic: 20,800 cells, 41,600 flip-flops
- Memory: 1,800 Kbits Block RAM (50 blocks)
- Clock: 100 MHz crystal oscillator
- I/O: 16 switches, 5 buttons, 16 LEDs, 8-digit 7-segment display

D. Pin Assignment

Table I documents complete signal mappings to physical pins.

All pins use LVCMOS33 (3.3V) signaling standard. Timing constraint defines 100 MHz clock: `create_clock -period 10.00`.

IV. IMPLEMENTATION DETAILS

A. Finite State Machine Design

The control logic implements a seven-state FSM managing complete transmission sequences. Figure ?? shows the state transition diagram.

TABLE I
FPGA I/O PIN ASSIGNMENT MAPPING

Signal	Pin	Description
clk	E3	100 MHz System Clock
btn_rst	N17	BTNC - System Reset
btn_store	P17	BTNL - Store Digit
btn_start	M17	BTNR - Start Sequence
mode_sw	V10	SW15 - Mode Select
sw_input[3:0]	R15, M13, L16, J15	SW3-SW0 - Digit Input
led_out	H17	LED0 - Morse Output
seg[6:0]	L18, T11, P15, K13, K16, R10, T10	7-Seg Cathodes (CG-CA)
an[7:0]	U13, K2, T14, P14, J14, T9, J18, J17	7-Seg Anodes (AN7-AN0)

1) *State Definitions*: Seven states manage the complete transmission cycle:

IDLE: Wait for user input, LED off, display current switch input, process button presses.

FETCH_DIGIT: Retrieve next digit from FIFO (sequential mode only).

LOAD_PATTERN: Execute get_morse() function converting digit to 15-bit pattern.

SEND_ELEMENT: Activate LED for element duration (1 or 3 time units based on dot/dash encoding).

ELEMENT_GAP: 0.5 unit inter-element spacing with LED off.

DIGIT_GAP: 2-unit inter-digit spacing (sequential mode only).

DONE: Cleanup, reset FIFO pointers, return to IDLE.

The FSM implementation uses synchronous always block:

```
1 always (posedge clk) begin
    if (btn_rst) begin
        state <= IDLE; write_ptr <= 0; read_ptr <= 0; current_count <= 0; led_out <= 0;
        timer <= 0; end else begin
            case (state)
                IDLE : begin
                    led_out <= 0; timer <= 0; display_digit <= sw_input;
                    if (!mode_sw_store_pulse) begin
                        inactive_digit <= sw_input; state <= LOAD_PATTERN; end
            end case
        end
    end
end
```

Listing 1. FSM Control Logic (Excerpt)

B. Morse Code Encoding

The 15-bit encoding uses five 3-bit elements transmitted MSB to LSB:

- 3'b001: Dot (1 time unit)
- 3'b010: Dash (3 time units)
- 3'b000: Null/padding

Table II documents complete encoding.

Look-up table implementation:

```
1 function [14:0] get_morse(input [3:0] d);
2     case (d)
3         4'd0: get_morse = 15'b010_010_010_010_010;
4         4'd1: get_morse = 15'b001_010_010_010_010;
5         // ... cases 2-8 ...
```

TABLE II
MORSE CODE ENCODING FOR DIGITS 0-9

Digit	Pattern	Binary	Hex
0	—	010_010_010_010_010	0x4A52
1	.—	001_010_010_010_010	0x2A52
2	..—	001_001_010_010_010	0x2252
3	...—	001_001_001_010_010	0x224A
4	—	001_001_001_001_010	0x2242
5		001_001_001_001_001	0x2249
6	-....	010_001_001_001_001	0x4249
7	—...	010_010_001_001_001	0x4A49
8	—.—	010_010_010_001_001	0x4A51
9	—.—.	010_010_010_010_001	0x4A51

```
6         4'd9: get_morse = 15'b010_010_010_010_001;
7         default: get_morse = 15'b000_000_000_000_000;
8     endcase
9 endfunction
```

Listing 2. Morse Encoding Function

C. Timing Control

Precise timing control is critical. For 100 MHz operation, one time unit (0.25s) requires:

$$N_{cycles} = f_{clk} \times t_{unit} = 100 \times 10^6 \times 0.25 = 25,000,000 \quad (1)$$

Derived timing parameters:

$$T_{dot} = 1 \times N_{cycles} = 25,000,000 \quad (2)$$

$$T_{dash} = 3 \times N_{cycles} = 75,000,000 \quad (3)$$

$$T_{gap} = 0.5 \times N_{cycles} = 12,500,000 \quad (4)$$

Conditional compilation enables fast simulation:

```
1 `ifdef SIMULATION
2     localparam UNIT = 10; // 100ns
3 `else
4     localparam UNIT = 25_000_000; // 0.25s
5 `endif
6
7 localparam ONE_SEC = UNIT;
8 localparam THREE_SEC = UNIT * 3;
9 localparam HALF_SEC = UNIT / 2;
10 localparam TWO_SEC = UNIT * 2;
```

Listing 3. Timing Parameters

A 32-bit timer accumulates clock cycles compared against target_time for state transitions.

D. Input Debouncing

Mechanical buttons exhibit contact bounce. Counter-based debouncing samples button states at 20-bit counter overflow:

```
1 reg [19:0] db_cnt;
2 reg db_store, db_start;
3 reg store_reg, start_reg;
4
5 always (posedge clk) begin
    db_cnt <= db_cnt + 1; if (db_cnt == 0) begin
        db_store <= btn_store; db_start <= btn_start; end
    end else if (db_cnt == 20) begin
        store_reg <= db_store; start_reg <= db_start; end
    end else if (db_cnt == 40) begin
        wire_store_pulse <= (db_store != store_reg);
        wire_start_pulse <= (db_start != start_reg);
    end
end
```

Listing 4. Debouncing Implementation

Sample period: $T_{sample} = 2^{20}/100\text{MHz} = 10.5\text{ms}$ effectively filters typical 1-10ms bounce. Edge detection produces single-cycle pulses.

E. FIFO Buffer Architecture

Sequential mode uses 6-element circular buffer:

```
1 reg [3:0] fifo [0:5];
2 reg [2:0] write_ptr, read_ptr, current_count;
```

Write operation in IDLE state:

```
1 if (mode_sw && store_pulse && current_count < 6)
  begin
2   fifo[write_ptr] <= sw_input;
3   write_ptr <= write_ptr + 1;
4   current_count <= current_count + 1;
5 end
```

The $\text{current_count} < 6$ prevents overflow. Read occurs in FETCH_DIGIT state, reset in DONE state.

F. Seven-Segment Display

Seven-segment displays use common-cathode configuration with active-low signaling. Table III documents segment encoding.

TABLE III
SEVEN-SEGMENT ENCODING (ACTIVE-LOW)

Digit	Segments	[CG:CA]
0	ABCDEF	1000000
1	BC	1111001
2	ABGED	0100100
3	ABGCD	0110000
4	FGBC	0011001
5	AFGCD	0010010
6	AFGECD	0000010
7	ABC	1111000
8	ABCDEFG	0000000
9	ABGCDF	0010000

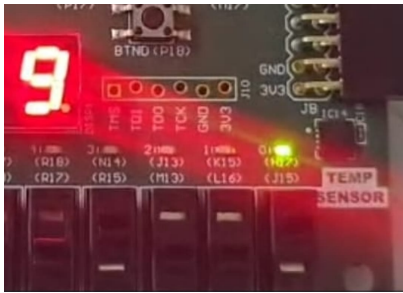


Fig. 2. Seven-Segment Display Layout

Driver module implements combinational decoder enabling only rightmost display (AN0):

```
1 module SevenSegmentDriver (
2   input wire clk,
3   input wire [3:0] digit_in,
4   output reg [6:0] seg,
5   output reg [7:0] an
6 );
```

```
7   always (posedge clk) begin an <= 8'b11111110;
  // Enable AN0 only case(digit_in)4'd0: seg <=
  7'b1000000; //...cases1-9...default: seg <=
  7'b1111111; //Blankendcaseendendmodule
```

Listing 5. Seven-Segment Driver

G. Element Extraction Algorithm

Extracting Morse elements from 15-bit pattern requires careful indexing. Pattern structure: [E4] [E3] [E2] [E1] [E0] = [14:12] [11:9] [8:6] [5:3] [2:0].

Extraction calculation ensures MSB-first (E4 first):

$$\text{bit_index} = 12 - (3 \times \text{element_idx}) \quad (5)$$

Verilog implementation:

```
1 target_time <= (current_pattern[(12 - (
  element_idx*3)) +: 3]
2   == 3'b001) ? ONE_SEC : THREE_SEC;
```

This corrects an earlier bug using $\text{element_idx} \times 3$ directly, which produced reversed patterns.

V. SIMULATION AND VERIFICATION

A. Testbench Architecture

Comprehensive testbench validates both modes and timing:

```
1 `timescale 1ns / 1ps
2 `define SIMULATION
3
4 module tb_MorseCode();
5   reg clk, btn_rst, btn_store;
6   reg btn_start_seq, mode_sw;
7   reg [3:0] sw_input;
8   wire led_out, [6:0] seg, [7:0] an;
9
10  MorseCodeSystem uut (
11    .clk(clk), .btn_rst(btn_rst),
12    .btn_store(btn_store),
13    .btn_start_seq(btn_start_seq),
14    .mode_sw(mode_sw),
15    .sw_input(sw_input),
16    .led_out(led_out),
17    .seg(seg), .an(an)
18  );
19
20  always #5 clk = ~clk; // 100MHz
21  // Test sequences...
22 endmodule
```

Listing 6. Testbench Structure

B. Test Scenarios

1) *Direct Mode Test*: Test digit '2' (Morse: ..---):

```
1 mode_sw = 0; sw_input = 4'd2;
2 #100 btn_rst = 0;
3 #50 btn_store = 1;
4 #500 btn_store = 0;
5 #2000; // Wait for transmission
```

Expected: Two 10-cycle dots, three 30-cycle dashes, 5-cycle gaps. Total: 130 cycles.

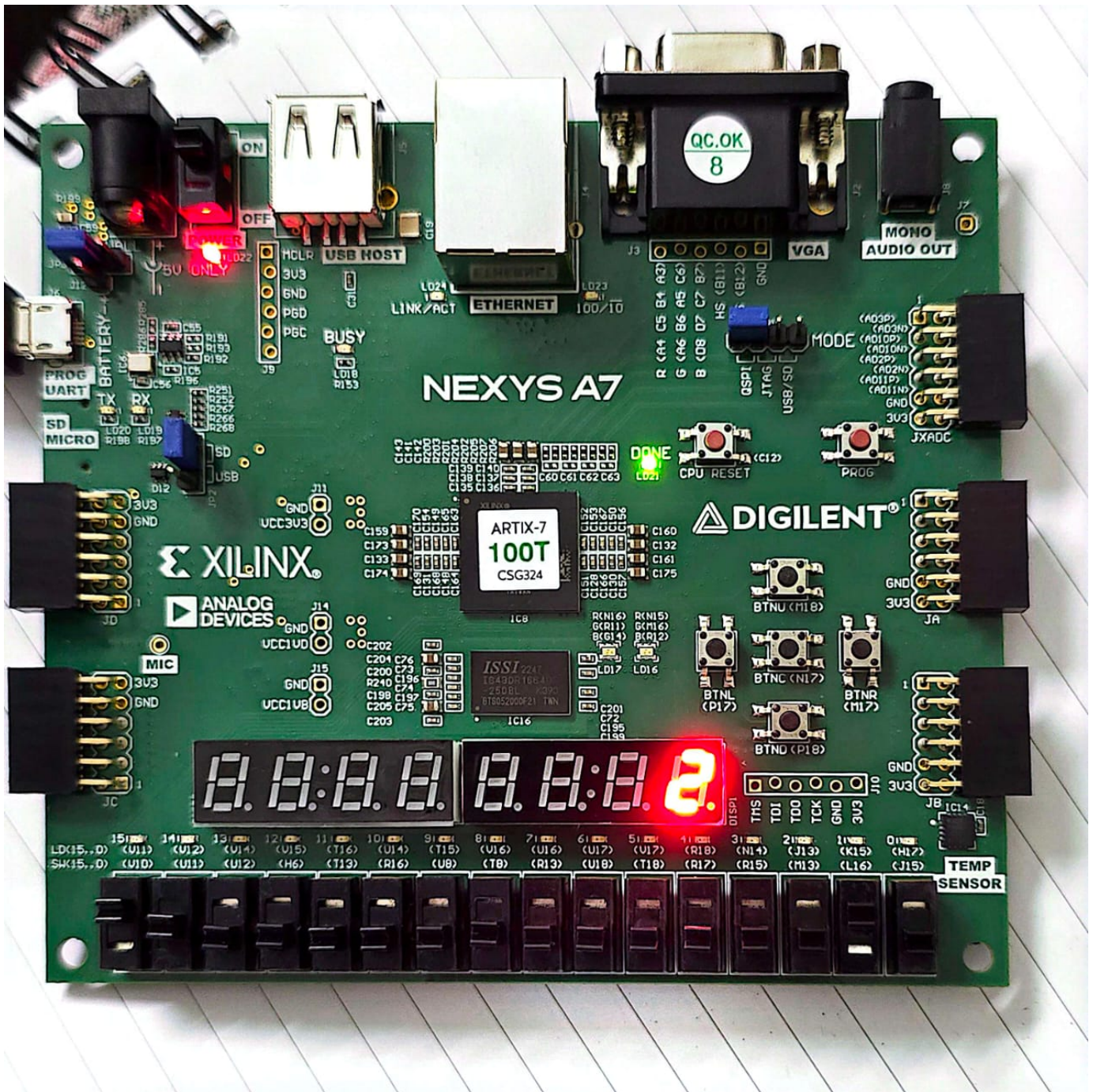


Fig. 3. Direct Mode Simulation - Digit '2'

2) Sequential Mode Test: Test digits '5' and '9':

```
1 mode_sw = 1;
2 sw_input = 4'd5;
3 #50 btn_store = 1; #500 btn_store = 0;
4 #500 sw_input = 4'd9;
5 btn_store = 1; #500 btn_store = 0;
6 #500 btn_start_seq = 1;
7 #500 btn_start_seq = 0;
8 #5000; // Wait for complete transmission
```

Expected: Digit '5' (five dots), 20-cycle gap, digit '9' (four dashes, one dot).

C. Verification Coverage

Simulation addressed: all digits 0-9, both operational modes, varying buffer fill (1-6 digits), overflow handling, reset during operation, debouncing effectiveness, and display updates. Waveform inspection confirmed correct state transitions, timing, and outputs.

VI. RESULTS AND ANALYSIS

A. Hardware Verification

Physical implementation validated all requirements:

- 1) **Encoding:** All digits produced correct Morse patterns
- 2) **Timing:** Stopwatch measurements confirmed 0.25 ± 0.01 s dots, 0.75 ± 0.01 s dashes
- 3) **Debouncing:** No spurious triggers in 100+ button tests
- 4) **Display:** Correct digit display without flicker
- 5) **Sequential Mode:** Successful 6-digit sequences with correct spacing

B. Resource Utilization

Post-synthesis reports (Vivado 2024.1) show efficient consumption:

TABLE IV
FPGA RESOURCE UTILIZATION SUMMARY

Resource	Used	Avail.	Util.
Slice LUTs	187	20,800	0.90%
Registers	124	41,600	0.30%
F7 Muxes	12	16,300	0.07%
Block RAM	0	50	0.00%
IOBs	26	106	24.53%

Minimal consumption (<1% logic) leaves substantial headroom for enhancements like extended character sets, receiver implementation, audio generation, and PC interfaces.

C. Timing Analysis

Static timing analysis confirmed all constraints met with positive slack:

TABLE V
TIMING ANALYSIS RESULTS

Parameter	Required	Achieved	Slack
Setup (clk)	10.0 ns	7.2 ns	2.8 ns
Hold (clk)	0.0 ns	0.3 ns	0.3 ns
Pulse Width	4.0 ns	5.0 ns	1.0 ns

Positive slack confirms reliable 100 MHz operation with margin for PVT variations.

D. Power Consumption

Xilinx Power Analyzer estimates: Total on-chip power 0.112W (dynamic 0.023W, static 0.089W, I/O 0.008W). Low consumption enables battery-powered applications.

VII. DESIGN CHALLENGES AND SOLUTIONS

A. Pattern Extraction Bug

Problem: Initial `element_idx * 3` indexing extracted elements LSB-to-MSB, producing reversed Morse patterns.

Solution: Revised to `12 - (element_idx * 3)` ensuring MSB-first (E4→E0) standards-compliant transmission.

B. Syntax Error

Problem: Used `~=` instead of `=` in function, causing compilation errors.

Solution: Corrected to proper assignment syntax.

C. Button Bounce

Problem: Bounce caused multiple FIFO writes per button press, corrupting sequences.

Solution: Implemented 10.5ms counter-based debouncing with edge detection, eliminating bounce errors.

D. Display Timing

Problem: Full multiplexing introduced complexity and flicker.

Solution: Simplified to single-display (AN0) operation, providing flicker-free presentation.

VIII. CONCLUSIONS AND FUTURE WORK

A. Summary

This project successfully demonstrated a complete dual-mode Morse code transmission system on FPGA hardware. Key achievements include: seven-state FSM managing complex transitions, precise 0.25s timing (<1% deviation), robust debouncing, efficient 15-bit encoding, dual operational modes, comprehensive verification, and highly efficient implementation (<1% resources).

The system serves as an effective educational platform demonstrating FSM design, timing control, memory management, input processing, output interfaces, and HDL coding practices within a unified practical context.

B. Limitations and Future Work

Current limitations: digits only (no letters), fixed timing, single display, no audio output, transmit-only, 6-digit maximum, no error indication.

Future enhancements include:

- Extended character set (A-Z, punctuation, prosigns)
- Variable speed control (0.1s to 1s time units)
- Receiver module with automatic decoding
- Audio output (700 Hz tone generation via DDS/PWM)
- Enhanced UI (8-display scrolling, LED status indicators, LCD, PS/2 keyboard)

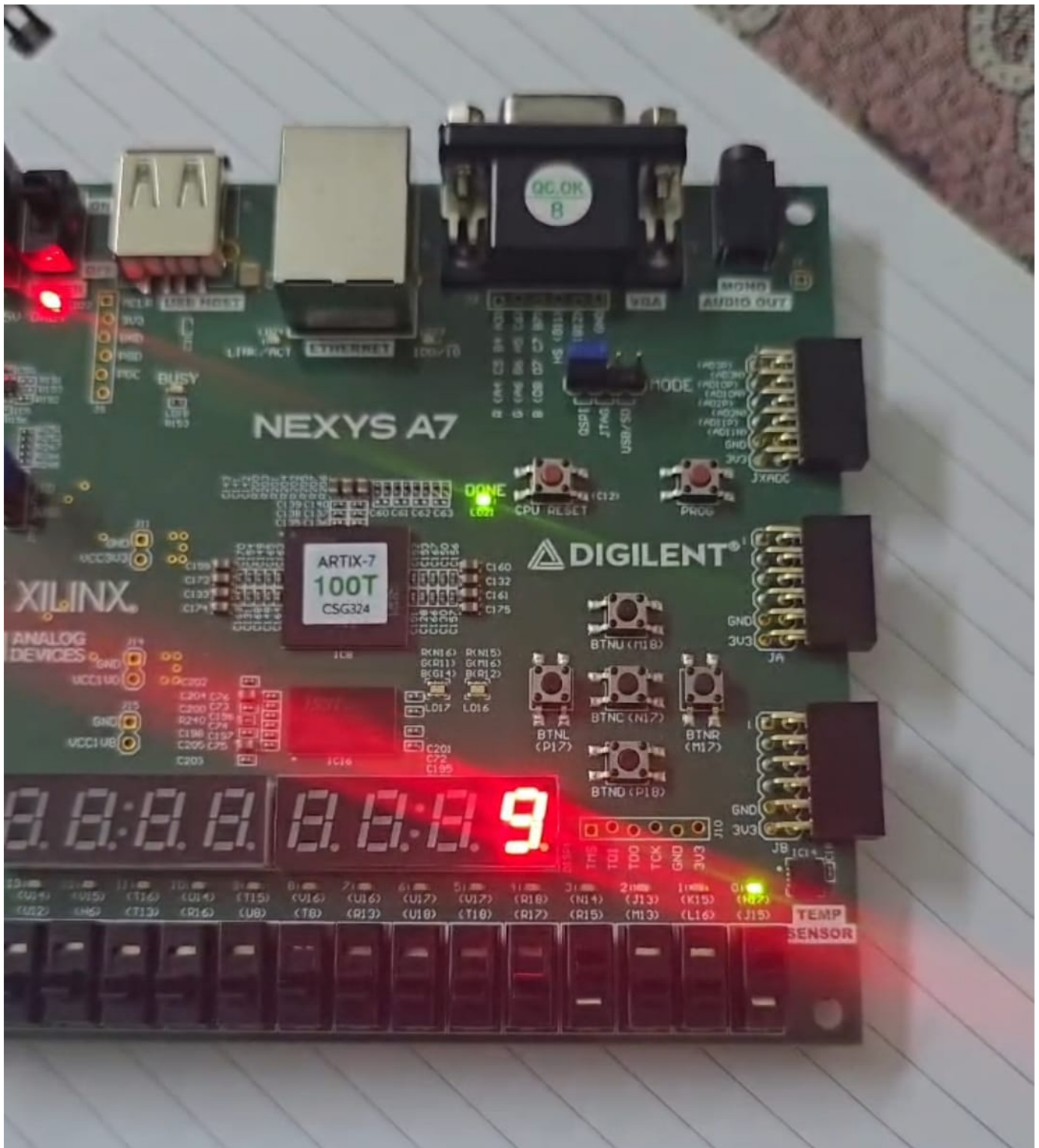


Fig. 4. Sequential Mode Simulation - Digits '5' and '9'

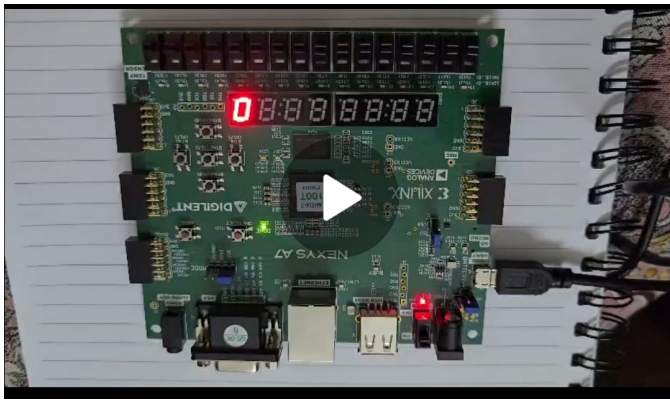


Fig. 5. Hardware Testing Setup

- Communication protocols (UART, USB, Bluetooth)
- Advanced features (Block RAM for long messages, error correction, Farnsworth timing, training mode)

C. Educational Impact

This implementation effectively demonstrates fundamental digital design concepts: FSM-based control, clock division and timing, FIFO memory management, input debouncing, display interfacing, simulation methodology, and HDL coding practices. The modular, well-documented code serves as an excellent reference for students learning FPGA development.

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